EJ'S BASE PRIME MODEL AND THE TWIN PRIME COUNTERFLOW MODEL: END-DIGIT KEYS, INNER-NODE SEQUENCES, AND TWIN-PRIME SEGMENT JUMPS

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Abstract. We develop EJ's Base Prime Model, a framework for primes greater than 5 that represents each prime by its terminal digit $d \in \{1, 3, 7, 9\}$ together with the preceding digits as a subscripted inner-node value. Gaps are denoted by a tilde prefix (e.g., $2 \sim$). We prove that the terminal digit of every gap is completely determined by the ordered pair of terminal digits of the bounding primes (Theorem 3.1), yielding the gap-group residues $\{0, 2, 4, 6, 8\}$. We further prove that every twin-prime pair greater than 5 realizes one of exactly three residue pairs modulo 9 (Theorem 10.2). Segments in the Base Model begin at the prime immediately preceding their interior and end immediately after a $2 \sim$; only the interior primes belong to the segment. The absolute value of any subscript is irrelevant; only the relative (normalized) sequence of subscripts inside a segment, beginning at 0, is conjectured to determine the exact sequence of intervening gaps. Individual primes act as short surface jumps, while twin-prime segments function as long-range jumps. Additional interpretive models (Twin Prime Counterflow via shared bg/ep boundaries, a Ladder visualization of matched subtractions, a Prime Scale musical mapping, a Lagrange-point analogy, and discrete-wave interpretations of empirical difference tables) are developed as correlated but independent perspectives on the same modular constraints. All density and sequence claims beyond the two modular theorems remain conjectural.

1. Introduction

Prime gaps have been studied extensively, yet local regularities based on terminal digits remain under-utilized for visualization and pattern extraction. EJ's Base Prime Model supplies a compact notation and a structural interpretation in which the leading digits of a prime form an "inner node" and the prime itself is a short jump connecting these nodes. Twin-prime pairs (gap exactly 2) and larger clusters of such pairs are interpreted as high-velocity conduits capable of bridging substantial composite intervals.

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The terminal-digit rule for gaps is elementary modular arithmetic and is proved in full. The deeper claim that ordered sequences of inner-node values determine gap structure inside segments is supported by extensive computation and is stated as a conjecture.

2. Notation

Definition 2.1. For a prime $p > 5$ write $p = 10n + d$ where $d \in \{1, 3, 7, 9\}$ and n is the non-negative integer formed by all preceding digits. The model notation is d_n. Gaps between consecutive primes are written $g\sim$ where g is the positive even integer gap.

Example 2.2. The initial segment beginning at 11, written in model notation, is

$$\begin{aligned} &1_1 \;\; 2\sim \;\; 3_1 \;\; 4\sim \;\; 7_1 \;\; 2\sim \;\; 9_1 \\ &\quad\; 4\sim \;\; 3_2 \;\; 6\sim \;\; 9_2 \;\; 2\sim \;\; 1_3 \;\; 6\sim \;\; \dots \end{aligned}$$

Here the terminal digits are $1, 3, 7, 9, 3, 9, 1$ and the successive gaps are $2\sim, 4\sim, 2\sim, 4\sim, 6\sim, 2\sim, 6\sim$. The interior primes of the first complete segment (after the exterior boundary 11 and before the twin ending at 19) have subscripts $1, 1, 1, 1$. Normalizing by subtracting the first subscript yields the relative sequence

$$(0, 0, 0, 0),$$

which is accompanied by the gap signature $2\sim, 4\sim, 2\sim$. The absolute values of the subscripts are irrelevant once this relative sequence is fixed; only the offsets from the first interior subscript enter the correspondence of Conjecture 4.3.

A second early segment beginning near 29 has interior primes whose subscripts normalize to $(0, 1, 1, 1)$ and produce the distinct gap signature $4\sim, 6\sim, 2\sim$. These two examples already illustrate that identical normalized sequences yield identical gaps while distinct sequences yield distinct gaps, exactly as claimed.

3. The Set Keys

Theorem 3.1. *Let p and q be consecutive primes greater than 5 with $p \equiv d_1 \pmod{10}$ and $q \equiv d_2 \pmod{10}$, where $d_1, d_2 \in \{1, 3, 7, 9\}$. Then the gap $g = q - p$ satisfies*

$$g \equiv d_2 - d_1 \pmod{10}.$$

The only possible residues are $0, 2, 4, 6, 8$. A residue of 0 corresponds to gaps belonging to the group $10, 20, 30, \dots$.

Proof. The congruence $g \equiv d_2 - d_1 \pmod{10}$ holds for any pair of integers ending in the indicated digits. Both d_1 and d_2 are odd, so their difference is even and the only admissible residues modulo 10 are $0, 2, 4, 6, 8$. Direct enumeration of the sixteen ordered pairs confirms uniqueness of the residue

for each pair (Table 1). The identity is therefore unconditional for all primes greater than 5. □

TABLE 1. Terminal digit of the gap (mod 10) for every ordered pair of terminal digits.

From \ To	1	3	7	9
1	0	2	6	8
3	8	0	4	6
7	4	6	0	2
9	2	4	8	0

These five residues are the gap-group labels $2\sim$, $4\sim$, $6\sim$, $8\sim$, or $10\sim$ (and multiples).

4. Inner Nodes and Segments

The subscript n attached to the terminal digit of a prime (the number formed by all preceding digits) is called the *inner node*. In EJ's Base Prime Model the absolute numerical value of any individual subscript is irrelevant. What matters is the ordered sequence of subscripts that appears inside a single segment. When that sequence is converted by subtracting the first subscript of the segment from every subsequent subscript, the resulting relative sequence begins at 0. This normalized sequence correlates exactly with the sequence of gap numbers that occur inside the segment.

Definition 4.1. A *segment* in EJ's Base Prime Model begins at the prime immediately preceding the interior primes of the segment and ends immediately after a gap of $2\sim$. Only the primes strictly inside these boundaries are considered part of the segment proper. The beginning prime (bg) of the current segment is the end prime of the previous segment, and the end prime (ep) of the current segment is the beginning prime of the next segment.

This strict boundary convention is specific to the Base Prime Model. The Counterflow Model, the Ladder Model, the Prime Scale Model, and the Lagrange-point analogy use the inversion that occurs at each $2\sim$ more freely for illustrative purposes; they do not require the same exterior-boundary accounting.

Definition 4.2. Let the interior primes of a segment have subscripts $n_1, n_2, \ldots, n_k$. The *normalized subscript sequence* of the segment is

$$(0,\ n_2 - n_1,\ n_3 - n_1,\ \ldots,\ n_k - n_1).$$

The first entry is always 0 by construction.

Conjecture 4.3. Inside any segment the normalized subscript sequence uniquely determines the exact sequence of gap numbers that appear between the interior primes. Distinct normalized sequences produce distinct gap signatures; identical normalized sequences produce identical gap signatures.

Thus the absolute size of the subscripts may be discarded once the segment boundaries are fixed; only the relative offsets from the first interior subscript remain, and those offsets stand in one-to-one correspondence with the intervening gaps.

4.1. **Equations of EJ's Base Prime Pattern.** Let the interior primes of a segment be $p_1, p_2, \ldots, p_k$ ($k \geq 2$) with corresponding subscripts $n_1, n_2, \ldots, n_k$. The absolute values n_i are discarded by the normalization map

$$N(n_1, \ldots, n_k) = \big(0,\ n_2 - n_1,\ n_3 - n_1,\ \ldots,\ n_k - n_1\big). \tag{4.1}$$

The first coordinate is identically zero. The Base Prime Pattern asserts that the normalized tuple uniquely determines the gap signature

$$\big(g_1 \sim,\ g_2 \sim,\ \ldots,\ g_{k-1} \sim\big), \tag{4.2}$$

where each gap is the ordinary difference $g_j = p_{j+1} - p_j$. In functional form there exists a map

$$\Phi\colon \big\{\text{normalized sequences}\big\} \to \big\{\text{gap signatures}\big\} \tag{4.3}$$

satisfying

$$\Phi\big(N(n_1, \ldots, n_k)\big) = \big(g_1,\ g_2,\ \ldots,\ g_{k-1}\big). \tag{4.4}$$

The modular end-digit keys of Theorem 3.1 constrain the image of Φ: every admissible gap residue must lie in $\{0, 2, 4, 6, 8\}$.

For the concrete segment whose interior primes are $11, 13, 17, 19$ one has subscripts $1, 1, 1, 1$, so

$$N(1, 1, 1, 1) = (0, 0, 0, 0)$$

and

$$\Phi(0, 0, 0, 0) = (2, 4, 2),$$

recovering the observed gaps $2 \sim, 4 \sim, 2 \sim$. The same normalized input arising from any later decade yields the identical gap signature, confirming that only the relative offsets enter Φ.

The table exhibits the precise correlation claimed in Conjecture 4.3: each distinct normalized sequence is accompanied by its own invariant gap signature, and identical normalized sequences (the two instances of $(0, 0, 0, 0)$) produce identical gap signatures. Additional segments drawn from larger ranges continue to obey the same rule.

Computational surveys of the first several hundred primes exhibit recurrent normalized motifs with invariant gap signatures. Representative early examples include the normalized sequence $(0, 0, 0, 0)$ (gaps $2 \sim, 4 \sim, 2 \sim$) and the normalized pairs $(0, 1)$ or $(0, 0)$ that accompany the decade transitions $3_n{-}9_n$ or $1_n{-}7_n$.

TABLE 2. Normalized subscript sequences and their exact gap signatures. The first entry of each normalized sequence is 0 by construction; the gaps listed are those that occur between the successive interior primes of the segment.

Segment (interior primes)	Normalized subscript sequence	Exact gap signature
191, 193, 197, 199	(0, 0, 0, 0)	$2\sim, 4\sim, 2\sim$
11, 13, 17, 19	(0, 0, 0, 0)	$2\sim, 4\sim, 2\sim$
127, 131, 137, 139	(0, 1, 1, 1)	$4\sim, 6\sim, 2\sim$
149, 151, 157, 163	(0, 1, 1, 2)	$2\sim, 6\sim, 6\sim$
223, 227, 229, 233	(0, 0, 0, 1)	$4\sim, 2\sim, 4\sim$
251, 257, 263, 269	(0, 0, 1, 1)	$6\sim, 6\sim, 6\sim$

5. Twin-Prime Segments as Long-Range Jumps

Individual primes function as short surface jumps between nearby inner nodes (gaps of 2, 4 or 6 typically remain inside or move only one decade). In contrast, a twin-prime pair (gap exactly $2\sim$) or a larger contiguous block of twin pairs acts as a high-velocity conduit. Once entered, the entire block may be treated as a single long-range jump that bridges many decades, analogous to the way a myelinated axon permits an action potential to skip large stretches of membrane.

Selected twin-prime pairs in the range 10–300, written in model notation, include

$$1_1\, 2\sim\, 3_1, \quad 7_1\, 2\sim\, 9_1, \quad 9_2\, 2\sim\, 1_3, \quad 1_4\, 2\sim\, 3_4, \quad 1_{10}\, 2\sim\, 3_{10}, \quad 7_{10}\, 2\sim\, 9_{10}.$$

Clusters of successive twin pairs (for example near 101–109 or 191–199) constitute especially long-range jumps.

Conjecture 5.1. Twin-prime segments of length $k \geq 2$ function as long-range jumps whose span (difference between the first and last inner node of the cluster) grows at least linearly with k.

6. The Twin Prime Counterflow Model

The Base Prime Model views the sequence of primes as a forward flow of short jumps between relative subscript offsets (normalized to begin at 0 inside each segment), with twin-prime segments acting as accelerated long-range conduits. The Twin Prime Counterflow Model supplies a dual perspective that does not rely on those subscript sequences at all.

From Theorem 3.1, the only terminal-digit pairs that produce a gap of exactly $2\sim$ are

$$(1,3), \qquad (7,9), \qquad (9,1).$$

These three transitions form a closed cycle under the modular keys. In the Counterflow Model a twin pair or a short twin-related cluster is treated as a

counter-current point at which the forward end-digit flow is locally balanced by the bg/ep relation.

Definition 6.1. Following Definition 4.1, a segment begins at the prime just before its interior and ends just after a 2 $\sim$. Only the interior primes belong to the segment. The bg of a segment is identified with the ep of the preceding segment, and the ep of a segment is identified with the bg of the succeeding segment.

Definition 6.2. A *counterflow balance* holds between consecutive segments when the subtraction of successive bg values exactly matches the subtraction of the corresponding ep values:

$$\mathrm{bg}_{i+1} - \mathrm{bg}_i = \mathrm{ep}_i - \mathrm{ep}_{i-1}.$$

Because bg and ep are shared with adjacent segments, the balance is independent of the particular interior content.

6.1. **Equations of the Twin Prime Counterflow Model.** Let successive segments have beginning primes bg_i and end primes ep_i. The Twin Prime Counterflow Pattern is expressed by the single balance equation

$$\mathrm{bg}_{i+1} - \mathrm{bg}_i = \mathrm{ep}_i - \mathrm{ep}_{i-1}. \tag{6.1}$$

Equivalently, the first differences of the bg sequence equal the first differences of the ep sequence (shifted by one index). In operator form,

$$\Delta \mathrm{bg}_i = \Delta \mathrm{ep}_{i-1}, \tag{6.2}$$

where $\Delta x_i = x_{i+1} - x_i$. The equation makes no reference to interior primes or to subscript sequences; it relates only the shared boundary values of adjacent segments. When the equality holds, the corresponding rung of the Ladder Model is balanced.

6.2. **List of Equations with Explanations.** The principal equations of the two models are collected and explained below.

(1) **Normalized subscript sequence (Base Prime Pattern).** The sequence
$$(0,\ n_2 - n_1,\ \ldots,\ n_k - n_1)$$
discards absolute subscript values. Subtracting the first interior subscript from every later subscript yields a relative sequence that always begins at 0. This normalized tuple is the sole input used by the Base Prime Pattern to predict gaps.

(2) **Gap-determining map (Base Prime Pattern).**
$$\Phi(0,\ n_2 - n_1,\ \ldots,\ n_k - n_1) = (g_1,\ g_2,\ \ldots,\ g_{k-1}).$$
The function Φ encodes the conjectured correspondence between normalized subscript sequences and exact gap signatures. Identical normalized inputs produce identical gap outputs; the modular end-digit keys constrain the possible values Φ may return.

(3) **Counterflow balance (Twin Prime Counterflow Pattern).**

$$\mathrm{bg}_{i+1} - \mathrm{bg}_i = \mathrm{ep}_i - \mathrm{ep}_{i-1}.$$

Successive beginning primes and successive end primes differ by the same amount. The equality relates only the shared boundary primes of adjacent segments.

(4) **First-difference form (Twin Prime Counterflow Pattern).**

$$\Delta\mathrm{bg}_i = \Delta\mathrm{ep}_{i-1},$$

where $\Delta x_i = x_{i+1} - x_i$. The same balance written with the forward-difference operator. When the equality holds, the corresponding ladder rung is balanced.

6.3. **End-Digit Restrictions on Boundary Primes.** In the end-digit pattern of the boundary primes themselves (the bg and ep values), the digit 3 appears as a beginning prime only once and thereafter appears only as an end prime after the first segment. Symmetrically, the digit 7 appears as an end prime only once and thereafter appears only as a beginning prime. These restrictions apply solely to the terminal digits of the shared boundary primes; they do not constrain the normalized subscript sequences inside the segments, nor do they constrain the gap numbers. The observation is compatible with the three admissible twin transitions and with the closed cycle $(1,3)$, $(7,9)$, $(9,1)$.

TABLE 3. Terminal digits of successive boundary primes illustrating the restriction. After the earliest segments, 3 appears only as an ep and 7 appears only as a bg.

Index	bg	bg end digit	ep end digit
0	11	1	9
1	29	9	1
2	41	1	3
3	59	9	1
4	71	1	3
5	101	1	9
6	137	7	9
7	149	9	1
8	179	9	1
9	191	1	3
10	197	7	9

6.4. **Back-Prediction via the Counterflow Ladder.** The Twin Prime Counterflow ladder supplies a reversible rule that recovers an ep difference from earlier bg differences. Consider three consecutive rungs indexed so that rung i has beginning primes $\mathrm{bg}_i, \mathrm{bg}_{i+1}$ (bottom left) and end primes

$\mathrm{ep}_i, \mathrm{ep}_{i+1}$ (top right). The back-prediction for the gap associated with rung $i+1$ is

$$(\mathrm{ep}_{i+1} - \mathrm{ep}_i) - (\mathrm{bg}_i - \mathrm{bg}_{i+1}). \tag{6.3}$$

The first term is the bottom-right difference; the second term is the top-left difference. Their difference is typically negative and equals the leading entry of the corresponding row in the scaled or offset columns of the Little Huffer table. That entry marks the start of the counterflow descent down and to the left along the ladder.

Example 6.3. Take the three rungs whose boundary primes are

$$\mathrm{bg}_1 = 29, \quad \mathrm{ep}_1 = 31,$$
$$\mathrm{bg}_2 = 41, \quad \mathrm{ep}_2 = 43,$$
$$\mathrm{bg}_3 = 59, \quad \mathrm{ep}_3 = 61.$$

Then

$$\mathrm{bg}_2 - \mathrm{bg}_3 = 41 - 59 = -18, \qquad \mathrm{ep}_3 - \mathrm{ep}_2 = 61 - 43 = 18,$$

and

$$(18) - (-18) = 36.$$

The value 36 appears as the leading scaled entry of the corresponding row in the Little Huffer table and initiates the counterflow descent. The same arithmetic applied to other triples recovers the offset-column entries, confirming that the ladder encodes a reversible prediction of successive ep gaps from bg differences alone.

The roles of the boundary primes are therefore asymmetric in the prediction step: the top-left (bg) pair supplies the quantity that is subtracted, while the bottom-right (ep) pair supplies the quantity from which it is subtracted. Interior primes and normalized subscript sequences remain unused.

Table 4 exhibits several twin-related segments and the independent bg/ep balances that hold for selected consecutive triples.

In the rows marked ✓the bg subtraction exactly equals the ep subtraction from the prior segment, demonstrating the independent balance. The Counterflow Model itself does not require an inner-node listing; the numerical matching between shared boundary primes is sufficient.

Conjecture 6.4. For every pair of consecutive segments defined by twin-prime anchors or by gaps of size at least 10, the counterflow balance

$$\mathrm{bg}_{i+1} - \mathrm{bg}_i = \mathrm{ep}_i - \mathrm{ep}_{i-1}$$

holds. The total gap of each segment remains consistent with the modular constraints, but the Counterflow Model itself does not invoke interior inner-node sequences.

The Counterflow Model therefore complements the Base Model by isolating the bidirectional numerical balance between beginning and end primes

TABLE 4. Twin-related segments and counterflow balances. The balance check compares successive bg subtractions with the corresponding ep subtractions; the equality is independent of any interior content.

Segment	bg	ep	Total gap	Balance check
S_0	11	19	8	—
S_1	29	31	2	$29-11=18$, $31-19=12$
S_2	41	43	2	$41-29=12$, $43-31=12$ ✓
S_3	59	61	2	$59-41=18$, $61-43=18$ ✓
S_4	71	73	2	$71-59=12$, $73-61=12$ ✓
S_5	101	109	8	$101-71=30$, $109-73=36$
S_6	137	139	2	$137-101=36$, $139-109=30$

of successive segments while remaining fully compatible with the modular end-digit keys. Inner nodes are not required for the balance itself.

7. The Ladder Model of Twin-Prime Sections

An auxiliary visual model arranges successive twin-prime sections into a ladder. Each rung of the ladder is a twin-related segment. On the bottom-left corner of each rung sit two beginning primes (bg values) whose mutual subtraction is recorded; on the top-right corner sit the two corresponding end primes (ep values) whose mutual subtraction is recorded. The defining property of the ladder is that these two subtractions match:

$$\mathrm{bg}_{i+1} - \mathrm{bg}_i = \mathrm{ep}_i - \mathrm{ep}_{i-1}.$$

The gaps between the rungs are thereby stylized: the vertical distance of each rung is the total gap of the segment, while the horizontal alignment of the matched subtractions makes the counterflow immediately visible as a series of equal-length cross-braces.

TABLE 5. Ladder of twin-prime sections. Each row records a pair of bg values (bottom left) and a pair of ep values (top right); the two differences are equal precisely when the rung is balanced.

Rung	bg pair (bottom left)	bg difference	ep pair (top right)	ep difference	Match
1	11, 29	18	19, 31	12	
2	29, 41	12	31, 43	12	✓
3	41, 59	18	43, 61	18	✓
4	59, 71	12	61, 73	12	✓
5	71, 101	30	73, 109	36	
6	101, 137	36	109, 139	30	

The ladder therefore supplies a purely visual counterpart of the Counterflow Model: the inner-node sequences that appear in the Base Prime Model are not needed here; only the shared boundary primes and their matched subtractions are required. The same construction can be repeated for the mod-9 residues or for the musical intervals, each time producing a ladder whose cross-braces are equal precisely when the underlying counterflow balance holds.

8. Musical Correlations: The Prime Scale and Harmonic Counterflow

The Twin Prime Counterflow Model admits a natural musical interpretation in which terminal digits function as pitch classes, gaps function as melodic intervals and rhythmic values, and twin-prime inversions function as melodic inversions that preserve harmonic content while reversing direction. This yields a distinct but correlated model—the Prime Scale Model—that maps the arithmetic structures onto harmony, melody, and rhythm.

8.1. **The Prime Scale.** Assign the four admissible terminal digits to the notes of a C-major seventh arpeggio (a stable harmonic skeleton):

$$1 \mapsto \mathrm{C},$$
$$3 \mapsto \mathrm{E},$$
$$7 \mapsto \mathrm{G},$$
$$9 \mapsto \mathrm{B}.$$

The resulting four-note collection $\{\mathrm{C}, \mathrm{E}, \mathrm{G}, \mathrm{B}\}$ is the *Prime Scale*. Every prime greater than 5 sounds one of these four pitches; the sequence of primes therefore traces a melody confined to this scale. Inner-node values $n = p//10$ act as octave registers or harmonic registers: larger n corresponds to higher octaves or more distant keys.

8.2. **Intervals, Keys, and Jump Sizes.** Gap magnitudes translate into melodic intervals and key-distance measures:

- Small base-prime jumps (gaps $2\sim, 4\sim, 6\sim$) correspond to stepwise motion or small leaps inside the home key of C.
- Medium twin-prime jumps (isolated $2\sim$ pairs) correspond to harmonic suspensions or brief neighboring tones that remain inside the same key.
- Large twin-prime segments connected by inversion at $2\sim$ (e.g., the pair 17–19 followed by its reverse contour 19–17) correspond to modulation or key change. The inversion $17\ 2\sim\ 19$ versus $19\ 2\sim\ 17$ is the melodic inversion of the interval G–B; both realize the same harmonic dyad $\{\mathrm{G}, \mathrm{B}\}$ while reversing direction, exactly as a musical inversion preserves interval class while reversing contour.

Key deviations arise precisely when the jump size increases. A short jump keeps the melody inside the Prime Scale of C; a medium jump may introduce a temporary neighboring pitch still related by the modular keys; a large jump (or a chain of inverted twin pairs) forces a change of register (inner node) and therefore a change of key, analogous to modulation to a closely related key.

8.3. **Rhythm and Harmony.** The sequence of gap sizes supplies rhythm: successive $2\sim$ gaps produce a steady pulse of eighth notes or quarter notes, while larger gaps ($8\sim, 10\sim, 14\sim$) introduce longer rhythmic values or rests. Harmony is generated by the simultaneous sounding of the three twin-prime transition pairs

$$(\mathrm{C},\mathrm{E}), \quad (\mathrm{G},\mathrm{B}), \quad (\mathrm{B},\mathrm{C}),$$

each of which is a consonant interval (major third, minor third, major second) inside the Prime Scale. The counterflow balance between bg and ep values corresponds to the conservation of harmonic tension across successive phrases: the numerical equality $\mathrm{bg}_{i+1} - \mathrm{bg}_i = \mathrm{ep}_i - \mathrm{ep}_{i-1}$ mirrors the conservation of interval sum across an inverted phrase.

TABLE 6. Mapping of Twin Prime Counterflow structures onto musical elements.

Arithmetic structure	Musical counterpart	Example
Terminal digit 1	Pitch C	$11 = 1_1$ sounds C
Terminal digit 3	Pitch E	$13 = 3_1$ sounds E
Terminal digit 7	Pitch G	$17 = 7_1$ sounds G
Terminal digit 9	Pitch B	$19 = 9_1$ sounds B
Gap $2\sim$	Stepwise motion / inversion	$17\ 2\sim\ 19$ (G–B) and its reverse
Gap $4\sim, 6\sim$	Small leap inside key	$13\ 4\sim\ 17$ (E–G)
Large gap $\geq 10\sim$	Modulation / key change	Jump from 19 to 23 changes register
Twin pair (1,3)	Major third C–E	Harmonic dyad
Twin pair (7,9)	Minor third G–B	Harmonic dyad
Twin pair (9,1)	Major second B–C	Harmonic dyad
bg/ep balance	Phrase-level harmonic conservation	$41 - 29 = 12$ matches $43 - 31 = 12$

Example 8.1 (Melodic inversion)**.** The twin pair $17\ 2\sim\ 19$ sounds G–B. Its contour inversion $19\ 2\sim\ 17$ sounds B–G. Both realize the identical harmonic interval (minor third) while reversing melodic direction—the precise musical analogue of the counterflow relation at gap $2\sim$.

Example 8.2 (Scale deviation and key change)**.** The short segment $1_1\ 2\sim\ 3_1\ 4\sim\ 7_1\ 2\sim\ 9_1$ sounds C–E–G–B and remains inside the home Prime Scale. The subsequent medium jump $9_1\ 4\sim\ 3_2$ moves to a new inner node and therefore to a new register/key, introducing the first scale deviation.

Larger chains of inverted twin pairs produce successive modulations whose distance is measured by the change in inner-node value.

The Prime Scale Model therefore supplies an independent but fully correlated language: every arithmetic relation proved or conjectured in the Base and Counterflow models translates into a corresponding statement about melody, harmony, or rhythm. The modular keys become the rules of voice-leading; the bg/ep balance becomes conservation of harmonic tension across inverted phrases; and the progression from small base jumps through medium twin jumps to large inverted twin segments becomes the progressive modulation from a home key into more distant keys.

9. Lagrange-Point Analogy: Earth, Lightning, and the Solar Inversion

The refined segment definition admits a further physical analogy drawn from celestial mechanics. Consider the Sun–Earth system and its Lagrange points.

- The Earth represents a reference prime or the start of an exterior boundary. - Lightning represents a short surface jump (a small gap) that carries the "light" (the prime progression) outward. - The Lagrange point lying behind the Sun corresponds to the gap $2\sim$: a balanced, unstable equilibrium point at which the forward flow is arrested. - Upon reaching this Lagrange point the trajectory inverts and falls into the Sun. The inversion is marked by the residue $0\sim$, and the destination is the special prime segment associated with 5 (the unique prime that ends in 5 and that initiates the first twin pair $5\ 2\sim\ 7$).

In this picture every segment is an orbital arc that begins just outside the previous boundary, travels via short jumps, reaches the Lagrange point $2\sim$, and then inverts into the central 5-segment via the $0\sim$ transition. The bg and ep of adjacent segments are the shared boundary points that keep the successive arcs in gravitational balance, exactly as the counterflow equality maintains numerical balance.

The analogy is consistent with the modular keys: the three twin transitions remain the only routes that can terminate at a $2\sim$ Lagrange point, and the subsequent inversion into the 5-segment is the unique place where a $0\sim$ residue appears in the early prime sequence.

10. The Mod 9 Twin Prime Counterflow Model

A parallel Counterflow Model is obtained by reducing every prime (the "rungs") modulo 9. Because the gap between twin primes is exactly 2, the residue of the second prime is completely determined by the residue of the first:

$$q \equiv p + 2 \pmod 9.$$

Primes greater than 3 can only be congruent to 1, 2, 4, 5, 7 or 8 modulo 9. The only transitions that preserve primality on both sides of a gap of 2 are

therefore

$$2 \to 4,$$
$$5 \to 7,$$
$$8 \to 1.$$

These three admissible mod-9 twin transitions form a closed cycle under addition of 2, exactly analogous to the three terminal-digit transitions $(1, 3)$, $(7, 9)$, $(9, 1)$ of the decimal model.

Definition 10.1. In the Mod 9 Twin Prime Counterflow Model each prime p is represented by its residue $r = p \bmod 9$. A twin-prime pair becomes one of the three rungs

$$2_9\ 2 \sim\ 4_9, \qquad 5_9\ 2 \sim\ 7_9, \qquad 8_9\ 2 \sim\ 1_9.$$

The inversion at the $2 \sim$ is again the point at which the forward residue flow reverses direction inside the mod-9 cycle.

TABLE 7. Mod-9 residues of the first twin-prime pairs and the corresponding counterflow transitions.

Twin pair	Residues mod 9	Transition	Type
11–13	2–4	$2 \to 4$	A
17–19	8–1	$8 \to 1$	C
29–31	2–4	$2 \to 4$	A
41–43	5–7	$5 \to 7$	B
59–61	5–7	$5 \to 7$	B
71–73	8–1	$8 \to 1$	C
101–103	2–4	$2 \to 4$	A
107–109	8–1	$8 \to 1$	C
137–139	2–4	$2 \to 4$	A
149–151	5–7	$5 \to 7$	B

The three types appear with roughly equal frequency, mirroring the equidistribution observed for the decimal end-digit pairs. Because the residue of every twin pair is forced by the modular arithmetic, the Mod 9 model supplies an independent verification of the same counterflow principle: the inversion at $2 \sim$ always lands on one of a closed set of three transitions, whether the underlying modulus is 10 or 9.

Theorem 10.2. *For any twin-prime pair p, q with $q = p + 2 > 5$, the residues $p \bmod 9$ and $q \bmod 9$ form one of the three ordered pairs $(2, 4)$, $(5, 7)$, or $(8, 1)$.*

Proof. The only residues modulo 9 that can be taken by a prime greater than 3 are 1, 2, 4, 5, 7, 8. Adding 2 modulo 9 to each admissible residue yields a second admissible residue if and only if the starting residue is 2, 5 or

8. Direct verification confirms the three pairs listed above; all other starting residues produce a multiple of 3, which cannot be prime. □

Conjecture 10.3. The ordered sequence of mod-9 counterflow types (A: $2 \to 4$, B: $5 \to 7$, C: $8 \to 1$) along successive twin-prime pairs is governed by the sequence of intervening inner-node values reduced modulo 9, and each type has asymptotic density one-third of the total twin-prime density.

The Mod 9 model is obtained simply by dividing the rungs by 9 (i.e., reducing modulo 9). It reproduces the same structural features as the original Counterflow Model—closed cycle of three transitions, inversion at each $2\sim$, and independent balance between successive segments—while remaining completely compatible with the terminal-digit keys of the Base Prime Model. The two modular pictures (mod 10 and mod 9) therefore supply mutually reinforcing descriptions of the same twin-prime counterflow.

11. Analysis of the Twin-Prime Flow Table (Little Huffer)

The accompanying spreadsheet tracks successive twin-prime groups by recording the starting prime of each twin pair, the next prime, the raw difference between consecutive twin-group starting numbers, a halved version of that difference, an "overlap" correction, and two offset versions of the difference sequence. All differences between twin-group starts are multiples of 6, as required by the form of twin primes greater than 3. The halved column and the scaled columns (frequently appearing as factors of 3) are consistent with the gap-group labels of the terminal-digit keys and with the mod-9 cycle.

The cumulative sums and offset columns implement a discrete version of the counterflow balance: each new difference is compared with a running total or with a shifted copy of the preceding differences. Where the offset column cancels or reduces the raw difference, the table is exhibiting the same numerical matching that the bg/ep relation isolates in the Counterflow Model. The presence of both positive and negative entries in the scaled columns further mirrors the inversion that occurs at each $2\sim$.

11.1. **Variations on the Flow Model.** Three immediate variations extend the same tracking procedure to the other models developed in this paper.

(1) **Mod-9 flow.** Replace each twin-start difference by its residue modulo 9. The resulting sequence takes values in $\{0, 3, 6\}$ and can be halved or scaled exactly as in the original table. The three admissible twin transitions of Theorem 10.2 appear as the only non-zero steps.
(2) **Musical-scale flow.** Map each twin-start difference onto the Prime Scale by the rule $d \mapsto$ the note corresponding to $d \bmod 4$ (or by the end-digit of the starting prime). Successive differences then become

melodic intervals; the offset column becomes a canonic inversion of the preceding phrase.

(3) **Inner-node flow.** Record the change in inner-node value between consecutive twin starts. The sequence of these changes is precisely the sequence whose cumulative sum equals the total span of the twin-prime segments; the offset version recovers the bg/ep balance of Definition 6.1.

Each variation preserves the modular constraints already proved (terminal-digit keys and mod-9 transitions) while making the counterflow visible as a running numerical or musical balance. The original table therefore supplies an empirical laboratory in which all of the models can be tracked simultaneously.

11.2. **Wave, Sine-Wave, and Refractive Behavior of the Highlighted Columns.** The scaled and offset columns of the Little Huffer table exhibit clear oscillatory structure. Entries frequently appear in proportional triples such as

$$1, 3, 6, \qquad -1, -3, -6, \qquad 3, 9, 18, \qquad -4, -12, -24, \qquad 4, 12, 24,$$

and similarly for larger magnitudes. The constant ratio 1:3:6 (or its negative) is precisely the scaling that converts a raw twin-group difference into the halved and overlap-corrected values already present in the table. When these triples are read in temporal order they form a discrete waveform whose amplitude is set by the underlying difference sequence and whose sign flips at regular intervals.

The sign reversals correspond to the inversion that occurs at each $2\sim$. In the language of the Counterflow Model the positive triples represent forward flow and the negative triples represent the counter-flow; their successive appearance produces a juggling or refractive effect in which the numerical "ray" is bent back upon itself exactly as light is refracted at an interface. The offset columns further modulate the waveform by subtracting a delayed copy of the preceding differences, generating the phase shift characteristic of a discrete sine-like oscillation.

A simple discrete sine model consistent with the observed data is

$$a_n = A \cdot \sin\Big(\frac{2\pi n}{T} + \phi\Big) \cdot s_n,$$

where A is the local amplitude taken from the halved difference column, T is a short period (commonly 4–6 steps, matching the length of many twin-related segments), ϕ is a phase fixed by the end-digit key of the starting prime, and $s_n = \pm 1$ encodes the counterflow inversion. The same construction works equally well for the mod-9 residues and for the musical-scale mapping: each admissible transition supplies a fixed phase increment, and the inversion at $2\sim$ supplies the sign change that keeps the wave bounded.

Thus the highlighted numbers do not merely tabulate differences; they generate a family of discrete waves whose amplitude, period, and phase are

controlled by the same modular keys, inner-node sequences, and counterflow inversions already developed in the paper. The refractive juggling visible in the sign changes is the concrete numerical counterpart of the Lagrange-point inversion and of the melodic inversion on the Prime Scale.

11.3. **Sine, Cosine, and Tangent Analogies.** The discrete wave generated by the scaled columns admits natural interpretations in terms of the elementary trigonometric functions, each of which corresponds to a distinct structural feature of the models.

- **Sine.** The fundamental oscillation is sine-like. Its argument is advanced by a phase increment fixed by the terminal-digit key (or the mod-9 transition) of the current twin start, while its amplitude is taken from the halved difference. The sign factor $s_n = \pm 1$ supplied by the inversion at each $2 \sim$ keeps the wave centered on zero, exactly as the counterflow balance keeps successive segments numerically matched. In the musical model the sine wave is the melodic contour itself; in the Lagrange analogy it is the radial oscillation about the equilibrium point.

- **Cosine.** Cosine is the phase-shifted companion of sine. In the present setting it corresponds to the offset columns of the Little Huffer table: subtracting a delayed copy of the preceding differences is equivalent to advancing the phase by $\pi/2$. The resulting cosine component tracks the rate of change of the sine wave and therefore measures how rapidly the counterflow is accumulating or discharging. In the segment language the cosine term records the difference between consecutive bg (or ep) values; in the musical language it is the harmonic "velocity" or the rate at which the melody modulates.

- **Tangent.** Tangent arises as the ratio of sine to cosine (or of the raw difference to its offset). Where the cosine component approaches zero the tangent becomes large, corresponding to those rare places where a large twin-group gap occurs and the local balance is momentarily lost. These spikes are precisely the large jumps that force a change of inner node or a modulation to a more distant key. In the Lagrange picture a near-zero cosine is the moment at which the trajectory passes closest to the unstable equilibrium and is then refracted sharply into the Sun. Thus tangent measures the local "sensitivity" of the counterflow and flags the transitions between short base-prime jumps and long-range twin segments.

A single plotted example (blue observed wave with sign flips, red sine approximation, green cosine) makes the three functions visible at once: the blue points follow the red sine when the counterflow sign is taken into account, while the green cosine tracks the discrete derivative and the large excursions of their ratio locate the refractive events.

The trigonometric triad therefore supplies a compact visual and analytic language that unifies the modular keys, the segment balance, the musical

contour, and the physical inversion. Extending the paper with further colored plots of the mod-9 residues, the inner-node changes, and the musical intervals will make each of these correspondences immediately readable.

12. Showcase of Patterns

An extended sequence of primes from 11 through 199, written in the model notation with gap markers, is

$$1_1\ 2\sim\ 3_1\ 4\sim\ 7_1\ 2\sim\ 9_1\ 4\sim\ 3_2\ 6\sim\ 9_2\ 2\sim\ 1_3\ 6\sim\ 7_3\ 4\sim$$
$$1_4\ 2\sim\ 3_4\ 4\sim\ 7_4\ 6\sim\ 3_5\ 6\sim\ 9_5\ 2\sim\ 1_6\ 6\sim\ 7_6\ 4\sim\ 1_7\ 2\sim$$
$$3_7\ 6\sim\ 9_7\ 4\sim\ 3_8\ 6\sim\ 9_8\ 8\sim\ 7_9\ 4\sim\ 1_{10}\ 2\sim\ 3_{10}\ 4\sim\ 7_{10}\ 2\sim$$
$$9_{10}\ 4\sim\ 3_{11}\ 14\sim\ \ldots\ 9_{19}.$$

Local motifs repeat with invariant gap signatures, supporting Conjecture 4.3.

13. Conclusion

EJ's Base Prime Model supplies a compact notation, a rigorously proved terminal-digit rule for gaps (Theorem 3.1), and a neural-network-inspired architecture in which primes are short surface jumps and twin-prime segments are long-range conduits. EJ's Base Prime Model supplies a compact notation, a rigorously proved terminal-digit rule for gaps (Theorem 3.1), and a refined segment definition in which each segment begins at the prime just before its interior and ends just after a 2 $\sim$, with bg and ep shared with adjacent segments. The Twin Prime Counterflow Model adds the bidirectional bg/ep balance. The musical correlation maps the same structures onto a Prime Scale, melodic inversion, and progressive modulation. The Mod 9 Counterflow Model, obtained by reducing the rungs modulo 9, yields a parallel closed cycle of three twin transitions and an independent theorem confirming that every twin pair realizes one of those three residues. Analysis of the accompanying twin-prime flow table shows that its difference, halved, and offset columns implement a concrete numerical version of the same counterflow; three variations of the table (mod-9, musical-scale, and inner-node) extend the tracking to every model developed above. The showcased sequences and tables demonstrate clear local regularities governed by the inner-node values. The combined models therefore offer both an immediately usable visual, computational, musical, physical, and modular language for prime-gap research and a set of concrete conjectures whose resolution would constitute a genuine advance in the understanding of the distribution of primes.

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